- Ass 3 due Fri 9 Oct @ 4pm

Recap:

① There is no decision rule that can simultaneously minimize Type I error & maximize power. There is always a trade-off between the two.

② The likelihood ratio test (LRT) that rejects $H_0: \theta = \theta_0$ in favour of $H_1: \theta = \theta_1$ if

$$\left\{ \frac{L(\theta_0 | \underline{x})}{L(\theta_1 | \underline{x})} \leq c \quad \text{for some critical value } c \right\}$$

is the MOST powerful test possible at any given level α (Neyman-Pearson lemma)

Example 7.3 (LRT for a normal mean)

Consider a random sample $X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ where we assume (for now) that σ^2 is known & only the mean μ is unknown

Suppose we are testing two point hypotheses about μ:

$H_0: \mu = \mu_0$

vs $H_1: \mu = \mu_1$, where μ_0 & μ_1 are two specified values.

Recall that the likelihood for for normal data:

$$L(\mu | \underline{x}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-1/2\sigma^2 \sum_{i=1}^{n} (x_i - \mu)^2}$$

So the LRT statistic is

$$\frac{L(\mu_0 | \underline{x})}{L(\mu_1 | \underline{x})} = \frac{e^{-1/2\sigma^2 \sum_{i=1}^{n} (x_i - \mu_0)^2}}{e^{-1/2\sigma^2 \sum_{i=1}^{n} (x_i - \mu_1)^2}}$$

$$= e^{1/2\sigma^2 \left[\sum_{i=1}^{n} (x_i - \mu_1)^2 - \sum_{i=1}^{n} (x_i - \mu_0)^2 \right]}$$

Small values of LR $\Leftrightarrow$ small values of $\sum_{i=1}^{n}(X_i-\mu_1)^2 - \sum_{i=1}^{n}(X_i-\mu_0)^2$

$\Leftrightarrow \sum_{i=1}^{n}(\underbrace{X_i-\bar{X}}_{a}+\underbrace{\bar{X}-\mu_1}_{b})^2 - \sum_{i=1}^{n}(\underbrace{X_i-\bar{X}}_{a}+\underbrace{\bar{X}-\mu_0}_{c})^2$

$= \sum_{i=1}^{n}(X_i-\bar{X})^2 + 2(\bar{X}-\mu_1)\sum_{i=1}^{n}(X_i-\bar{X}) + n(\bar{X}-\mu_1)^2$

(a^2; $2ba$; $\sum(X_i-\bar{X}) = 0$; b^2)

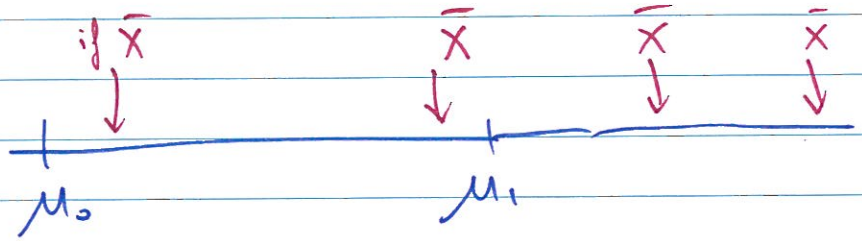
$- \sum_{i=1}^{n}(X_i-\bar{X})^2 - 2(\bar{X}-\mu_0)\sum_{i=1}^{n}(X_i-\bar{X}) - n(\bar{X}-\mu_0)^2$

(a^2; $2ca$; $\sum(X_i-\bar{X}) = 0$; c^2)

$= n\left[(\bar{X}-\mu_1)^2 - (\bar{X}-\mu_0)^2\right]$

$\Leftrightarrow$ Small values of $(\bar{X}-\mu_1)^2 - (\bar{X}-\mu_0)^2$ (Note: this depends only $\bar{X}$, which is sufficient for μ)

If alternative $\mu_1 > \mu_0$



$\Leftrightarrow$ large values of $\bar{X}$

So the LRT starts of as

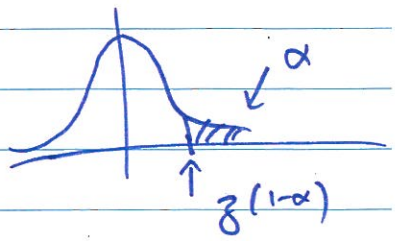
$$\left\{\frac{L(\mu_0|\underline{X})}{L(\mu_1|\underline{X})} \leq c \text{ for some } c\right\} \overset{\text{becomes}}{\Leftrightarrow} \left\{\bar{X} \geq d, \text{ for some } d\right\}$$

How do we set d?

If we want to control Type I error to at most some level α ($= 0.05$, say), then

$$\alpha = P(\bar{Z} \geq z^{(1-\alpha)})$$

$$\overset{(CLT)}{\approx} P_{H_0}\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq z^{(1-\alpha)}\right)$$



$$= P_{H_0}\left(\bar{X} \geq \underbrace{\mu_0 + z^{(1-\alpha)}\sigma/\sqrt{n}}_{d}\right)$$

For our medical example, $\mu_0 = 0$ (placebo), $\sigma = 1.1$ & $n = 27$. So for $\alpha = 0.05$, the LRT reduces to the test

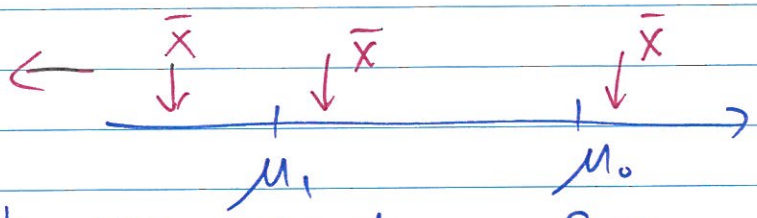
$$\left\{\bar{X} \geq 0 + 1.645 \times 1.1/\sqrt{27}\right\}$$

$$= \left\{\bar{X} \geq 0.348\right\} \text{ as before.}$$

This test is the most powerful test possible (at the level α) for deciding between $H_0: \mu = \mu_0$ vs $H_1: \mu = \mu_1 > \mu_0$.

Because this test does not explicitly depend on the value of μ_1 in the alternative, it is in fact the uniformly most powerful test (UMP) for $H_0: \mu = \mu_0$ versus $H_1: \mu > \mu_0$ (i.e. most powerful test for all μ_1 in the alternative).

Similarly, if we are testing $H_0: \mu = \mu_0$ vs $H_1: \mu = \mu_1 < \mu_0$ then small values of $(\bar{X} - \mu_1)^2 - (\bar{X} - \mu_0)^2$:



the LRT reduces to $\{\bar{X} \leq d, \text{ for some } d\}$.

($\Leftrightarrow$) if we specify a specific α, the test is explicitly

$$\{\bar{X} \leq \mu_0 - z^{(1-\alpha)}\sigma/\sqrt{n}\}$$

& this is UMP for $H_0: \mu = \mu_0$ vs $H_1: \mu < \mu_0$

If we are testing $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$, the LRT reduces to

$$\{\bar{X} \leq d_1 \quad \text{OR} \quad \bar{X} \geq d_2\}$$

$$\Leftrightarrow \{\bar{X} \leq \mu_0 - z^{(1-\alpha/2)}\sigma/\sqrt{n} \text{ OR } \bar{X} \geq \mu_0 + z^{(1-\alpha/2)}\sigma/\sqrt{n}\}$$

Note: $|z^{(1-\alpha/2)}| > |z^{(1-\alpha)}|$ (e.g. $1.96 > 1.645$)

So two-sided tests needs <u>more evidence</u> in either direction.

Is the two-sided test UMP? NO, two-sided tests cannot be UMP because a correctly chosen one-sided test will always be more powerful than a two-sided one.

We see that for normal* data, the LRT for testing μ reduces to large and/or small values of $\bar{X}$. To set our critical values for the test, we used the result

$$\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \overset{\text{approx}}{\sim} N(0,1) \quad \text{under } H_0: \mu=\mu_0$$

What if we did NOT know σ^2? We can use S^2 (sample variance) to estimate σ^2. Then the LRT reduces to large and/or small values of the pivot

$$\frac{\bar{X}-\mu_0}{S/\sqrt{n}} \overset{\text{approx}}{\sim} t_{(n-1)} \quad \text{under } H_0: \mu=\mu_0.$$

To control Type I of the test to at most α, we use the t-cutoffs instead of the z-cutoffs:

$$\left\{ \bar{X} \geq \mu_0 + t_{(n-1)}^{(1-\alpha)} S/\sqrt{n} \right\}$$

for testing $H_0: \mu=\mu_0$ vs $H_1: \mu>\mu_0$

$$\left\{ \bar{X} \leq \mu_0 - t_{(n-1)}^{(1-\alpha)} S/\sqrt{n} \right\}$$

for testing $H_0: \mu=\mu_0$ vs $H_1: \mu<\mu_0$

$$\left\{ \begin{array}{l} \bar{X} \leq \mu_0 - t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \\ \text{OR} \\ \bar{X} \geq \mu_0 + t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \end{array} \right\}$$

for testing $H_0: \mu=\mu_0$ vs $H_1: \mu\neq\mu_0$

This is how one-sided & two-sided t-test can be formally justified

Finally, just how critical is the normal* data assumption?
We see from tutorial 9 that t-methods are incredibly robust against deviations from normality.

Of course, t-tests are NOT most powerful if the data are non-normal, but they can still be asymptotically most powerful! That is, as the sample size increases, the t-test has power that approaches the correctly specified LRT.

LRT for Poisson data (parallel to Tutorial Week 10)

Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} \text{Poisson}(\lambda)$. Suppose we are testing between $H_0: \lambda = \lambda_0$ ($= 7.3$ calls/hr)
vs $H_1: \lambda = \lambda_1 > \lambda_0$

Construct a LRT for this scenario:
Recall the likelihood for for a Poisson is

$$L(\lambda; \underline{X}) = \prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{X_i}}{X_i!}$$

$$= e^{-n\lambda} \frac{\lambda^{\sum_{i=1}^{n} X_i}}{\prod_{i=1}^{n} X_i!}$$

So the LRT statistic is

$$\frac{L(\lambda_0 | \underline{X})}{L(\lambda_1 | \underline{X})} = \frac{e^{-n\lambda_0} \lambda_0^{\sum_{i=1}^{n} X_i}}{e^{-n\lambda_1} \lambda_1^{\sum_{i=1}^{n} X_i}}$$

$$= \underbrace{e^{n(\lambda_1 - \lambda_0)}}_{\text{always positive}} \left(\frac{\lambda_0}{\lambda_1}\right)^{\sum_{i=1}^{n} X_i}$$

< 1

Since $\lambda_1 > \lambda_0$, the LRT is small when

$$\sum_{i=1}^{n} X_i \text{ is large.}$$

So the LRT for testing $H_0: \lambda = \lambda_0$ vs $H_1: \lambda = \lambda_1 > \lambda_0$ reduces to a single rule

$$\left\{ \sum_{i=1}^{n} X_i \geq d, \text{ for some critical value } d \right\}$$

"total number (of calls) is too large"

Again, this rule will not depend on the value of λ_1 in the alternative, & so it is in the uniformly most powerful against any alternative ~~$\lambda_1 > \lambda_0$~~, i.e. $H_0: \lambda = \lambda_0$ vs $H_1: \lambda > \lambda_0$, for Poisson data.

How do we choose critical value d to control Type I error to at most level α ($= 0.05$, say).

In STAT2003 (see also Tutorial Week 10) that sums of independent Poissons is still a Poisson:

ie. $X_1 + X_2 + \cdots + X_n \sim \text{Pois}(n\lambda_0)$

So to control Type I error to α, we need to find d s.t.

$$\alpha \geq P_{H_0}(X_1 + X_2 + \cdots + X_n \geq d)$$
$$= P_{H_0}(\text{Pois}(n\lambda_0) \geq d)$$

& so d must be the upper α-percentile of a $\text{Poisson}(n\lambda_0)$

Suppose, we have $n=15$, $\lambda_0 = 7.3$ calls/hr
& setting $\alpha = 0.05$, we have

$$0.05 \geq P\left(\text{Pois}(15 \times 7.3) \geq d\right)$$
$$= P\left(\text{Pois}(109.5) \geq d\right)$$

start with q λ

```
> qpois(0.95, 109.5)
[1] 127
```

Check:

```
> 1 - ppois(127, 109.5)
[1] 0.04536

> 1 - ppois(126, 109.5)      "P(Y ≤ 126)"
[1] 0.05473
```

So our rule to reject $H_0: \lambda = 7.3$ calls/hr in favor of $H_1: \lambda > 7.3$ is

$$\{ X_1 + X_2 + \cdots + X_{15} \geq 128 \text{ calls} \}$$

$$\Leftrightarrow \{ \bar{X} \geq \frac{128}{15} = 8.53 \text{ calls/hr over } n=15 \text{ blocks} \}$$

Suppose we ignore the fact that #calls is Poisson, & we blindly use the t-test:

$$\{ \bar{X} \geq 7.3 + t_{14}^{(0.95)} S/\sqrt{n} \}$$

$$= \{ \bar{X} \geq 7.3 + 1.76\, S/\sqrt{n} \}$$

The are qualitatively & quantitatively two different rules!
Which is better in practice?

Type I error: both of these rules have been set to level (approx) 5% under $H_0: \lambda = \lambda_0$

Power: If data are truly Poisson, then Poisson LRT is most powerful. (Neyman-Pearson lemma). If data were NOT poisson, firstly Type I error may not be close to α & the power may also be much lower. In contrast, the t-test always has (approx). Type I error & asymptotically highest power!

Knowing the correct model (likelihood) of the data leads to the LRT as the optimal decision rule. However, in practice we never know the exact data-generating distn & so aiming for asymptotically optimal tests that are robust to model misspecification is important. (see STAT3500 for an intro to non-parametric tests)

Note that in our discussion of hypothesis testing so far, we have not looked at any numerical data yet. I have done this on purpose to emphasize that:

(i) identifying pop' parameter(s) of interest
(ii) formulating our null & alternative hypotheses about the parameter.
(iii) choosing the test statistic $T(\underline{X})$ & region $\{T(\underline{X}) \in C\}$
(iv) set Type I error of the test
(v) power assessment of our test

must be carried out before actually collecting numerical data. These are properties of the testing procedure & not specific to any sample of data.

Once we have carried out our experiment & collected our numerical data, we can then go ahead & make our decision based on our decision rule. We can also supplement our decision using a p-value, which quantifies strength of evidence against H_0 in favour of H_1.

p-values Suppose we have realisations $x_1, x_2, \ldots, x_n$ of $X_1, X_2, \ldots, X_n$ from some random experiment. We can then compute our test statistic

$$t_{obs} = T(\underline{x})$$ & make our decision based on this value.

Whether we choose to accept or reject H_0, we can supplement this decision by computing the p-value:

$$\text{p-value} = P_{H_0}(T(\underline{X}) \geq t_{obs}) \quad \text{for } H_0: \theta = \theta_0 \text{ vs } H_1: \theta > \theta_0$$

$$\text{p-value} = P_{H_0}(T(\underline{X}) \leq t_{obs}) \quad \text{for } H_0: \theta = \theta_0 \text{ vs } H_1: \theta < \theta_0$$

$$\text{p-value} = 2 \times \min \begin{cases} P_{H_0}(T(\underline{X}) \geq t_{obs}) \\ P_{H_0}(T(\underline{X}) \leq t_{obs}) \end{cases} \quad \text{for } H_0: \theta = \theta_0 \text{ vs } H_1: \theta \neq \theta_0$$

In all three cases, the p-value is always the probability under the null H_0 that our test statistic would be "as unusual or more unusual than what we observed".

Eg (Tutorial Week 10, continued). We have collected $n = 15$ Poisson counts, & the observed values are

7, 11, 10, 11, 6, 7, 10, 9, 7, 5, 11, 5, 9, 6, 18

we are testing for $H_0: \lambda = 7.3$ vs $H_1: \lambda > 7.3$.

(f) Recall our decision rule at the 5% level was

$$\left\{ \sum_{i=1}^{n} X_i \geq 128 \right\}.$$

After collecting our data our observed total was

$$7 + 11 + 10 + \cdots + 18 = 132 \text{ calls}$$

So based on our rule, we decide to reject H_0 in favour of H_1.

(g) Supplement your decision by computing & interpreting a p-value:

$$\text{p-value} = P_{H_0}\left(\sum_{i=1}^{15} X_i \geq 132 \right) \quad \leftarrow \text{obs. value}$$

$$= P_{H_0}\left(\text{Pois}(15 \times 7.3) \geq 132 \right)$$

$$= P_{H_0}\left(\text{Pois}(109.5) \geq 132 \right)$$

$$> 1 - \text{ppois}(131, 109.5)$$

$$\doteq 0.02 = 2\%$$

Interpret: If the null claim were true ($\lambda = 7.3$) then observing 132 calls or more from $n = 15$ hourly blocks would occur 2% of the time. This is quite rare, & so reasonably strong evidence against H_0 in favour of H_1.